CAS (Heimdall) – system functional description

Inlet system: water intake and pen level control

ScaleAQ Automation Document CAS-FD-10, rev. B5

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Abstract

How the inlet system of the CAS (Heimdall) closed fish pen works and how the control system controls it. Three Framo inlet pumps fill the pen with sea water; the Heimdall PLC holds the water inside the pen a few centimetres above the sea outside (the *over-height*) with one common pump speed, and chooses how many pumps run. The document covers every instrument and actuator, the control principle, sequences, the pump state machine, interlocks, alarms and faults. Undecided points are marked OIn and listed in Section 6.

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1 Introduction

1.1 Purpose and readers

The document is the system functional specification of the inlet system in the sense of [COO] §14.3: a description of the physical system (*block diagram description*), of what the control system does with it (*functional diagram description*) and of the functional and non-functional requirements. It is written for

- **engineers and the project team**, who implement, test and commission the PLC program;
- **operators**, who need to know what the system does on its own and what they must do;
- **suppliers** (Framo for the inlet pumps), who must agree the interface and the behaviour at the boundary.

[FD] (Norwegian) is authoritative on the *intent*. This document gives the intent in English, shows what the PLC program [PLC] implements today (main, 2026-10-22) and where they differ.

1.2 Scope

In scope: the water intake from the three intakes to the pen (Framo inlet pumps PM-31/32/33, their local controllers and vacuum systems as seen from the PLC), the level measurement inside and outside the pen, the level controller LIC01, the start and stop of the pen and of each pump, the communication with the Framo controllers and the related alarms.

Out of scope (separate FDs): the oxygen system (side stream, oxy pumps PU41–43, OIC01), the outlet dampers and their hydraulics, the bilge system in the inlet boxes, and the power distribution. They appear here only where they interface with the inlet system: the oxygenation permit per line, the level setpoint per operating mode from the outlet, and interlocks from the inlet boxes.

1.3 Sources

Table 1: Sources. The IDs are used in the text.

ID	Document	Revision / date
[FD]	P-15169-CAS-FD-01 *Funksjonsbeskrivelse CAS*, §3.2.1 Vanninntak, §3.2.2 level control, tables 5–10 (Norwegian)	rev. B, 13.05.2026
[AL]	P-15169-CAS-FD-02 *Alarmliste CAS Heimdall*	rev. A
[FRAMO]	Framo, *Spesifikasjon: kontrollsystem (prototype)* 0000_CAS_SPEC_CTRLSYS, signal list and state table, with ScaleAQ review comments	rev. B, 08.10.2025
[FIL]	Framo interface list (early signal list)	01.09.2025
[FRAMO-FD]	Framo, *Functional description* 7664-012-4, issued for review (no HOLD; link loss; stand-alone)	rev. 04, 05.11.2026
[FRAMO-SCD]	Framo, *System control diagram* 7664-017-1	rev. A03, 03.11.2026
[PID]	CAS P&ID and loop diagrams: CAS-PID-100 inlet, -400 pen, -600 utilities, CAS-LAY-700 network, CAS-LD-801 loop LT01, CAS-PID-950 open points	rev. A, 2026-10-16
[DES]	CAS design basis (System Modelling), generated from design_basis.toml	v2.3.12
[PDS]	CAS program design (dataflow of the level loop, start command path)	2026-10-15
[PLC]	Heimdall PLC program (CODESYS, application PFC200_Sim): FB.Inlet.InletControl, InletSupervisor, InletSensors, FB.Pen.PenControl, PRG.FramoDataProcessing, GVL ConfigValues	main, 2026-10-22
[EL]	Electrical drawings 15169-792.001 (PLC/RIO1 cabinet), 792.002 (RIO2/3 cabinet), 874.001 (Framo pump cabinet)	792.001 rev. 1, 792.002 rev. 0
[COO]	J. Cooling, *Software Engineering for Real-Time Systems*, 2020, §14.3 System functional specifications	2020

1.4 Abbreviations and terms

Term	Meaning
CAS	Closed Aquaculture System; Heimdall is the control system of the CAS pen
Over-height, ΔH	Water level inside the pen minus sea level outside [m or cm]
ΔH_i, pair i	Over-height measured at inlet i: LT0i − LT2i
PV, SP	Process value, setpoint
LIC01	Level indicating controller of the pen (PI)
Framo PCCn	Pump control cabinet of inlet n: Beckhoff CX8200 PLC and ABB ACS880 drive
PFC200	WAGO 750-8212, the Heimdall PLC
RIO	Remote I/O: WAGO 750-362 Modbus TCP couplers (CommonRIO, InletRIO1–3)
OPC UA	Communication between the PLC and the Framo controllers (and the HMI)
HMI	Operator interface (Heimdall HMI / SiteControl)
VSD	Variable speed drive
REMOTE, LINK LOST, ...	Pump states in the PLC, from the Framo controller's flags and the link (Section 3.5)
O-35, O-45, O-60, S-75	Operating modes: operation (O) or service (S), with residence time in minutes
TBD	To be decided; see the open issues

1.5 Revision note

Rev.	Date	Change
A	2026-10-21	First issue. Based on [FD] rev. B, [FRAMO] rev. B, [PID] rev. A and the PLC branch in review (heartbeat, HOLD, per-pair PV and SafeOutput changes of 2026-10-15). Includes the review and simulator test of 2026-10-21.
A1	2026-10-21	Reset to the PCC: held until the state change is seen, warning *Failed to reset* after 2×`ResetExpectedTime`; HOLD→REMOTE open issue reworded (the PLC follows the PCC either way); OI24 HMI Reset button.
B	2026-10-22	LIC01's output is the *total* speed of the pumps in level control, shared by the pumps in REMOTE (Section 3.2, DEC5; K_p 1350 total rpm/m, one parameter set; OI5, OI6 closed). InletRIO link loss drops that inlet's pair (DEC6, OI17 closed). Level deviation alarms high/low (DEC8, OI8 in part). Automatic pump count and rotation by run time with `PumpCycleTime` (Section 3.4, DEC7, OI15 in part). Speed deviation alarm (OI9); LowPower / HighPower out of the common alarm (OI10 closed). Live test 2026-10-22 of the start, sharing, pump count and rotation (Section 5). Operator commands restructured: pen state machine with Ready and pulses only, controller modes as pulses with a status, internal commands apart (Section 3.3). Simulator tests of the sharing, the pump count, the rotation and the pen states are pending.
B1	2026-10-23	LIC01 operator auto/manual bumpless both ways (IN-32): auto→manual starts the manual output at the current output per pump (before, it jumped to the last `ManualOutput`, 40 rpm if never set); manual→auto sets the setpoint to the PV and restarts with a cleared integral. Util.PID `RESET` / `Y\_OFFSET` use and the tuning-only-at-reset rule described (Section 3.2); the manual output is limited to the output range in `PIDController`. HOLD no longer freezes LIC01 (DEC9): rules R1/R2 replace tracking, freeze and resume; `InletSupervisor` split into `InletPumpSet`, `PenLevelControl`, `PumpCount`, `PenOxygenControl`, `PenSequence`, `ControllerMode` (IN-33: before, a pump reset from HOLD dropped all pumps to 45–49 rpm). The operator's manual output is per pump in REMOTE.
B2	2026-11-03	After the merge with the outlet / oxygen branch (InletPumpConfigDUT removed): the pump count flag is `Inlet[i].InletPump.Config.InletPumpEnable`. Alarms only from pumps in service, PCC flag alarms only with a live link, LIC01 / OIC01 limit and overflow alarms only in automatic control (Alarm functions). The split `InletSupervisor` of B1 restored after the merge had reverted it to the pre-B1 version. InletRIO freshness watchdog: a link whose master still reports connected but sends no new data for 2 s counts as lost, and only that master is restarted (fault table). Framo data sources: endpoint and update rate from the retained `Inlet[i].InletPump.Config` (dynamic configuration, `FramoLink`), reconnect on change or a lost link, link live only with the data source connected (Communication).
B3	2026-11-05	PLC node structure by system: the inlet station's remote I/O is `Inlet[i].RIO`, the pump's settings `Inlet[i].InletPump.Config`; LIC01 is `Pen.LevelController` with its settings in `.Config` and the setpoint in `.Command.Setpoint` (retained); the inlet supervisor and pump order are `Pen.InletSupervisor`. Unused nodes commented out in the PLC, among them the per-transmitter level alarms of `Inlet[i].Alarm` (never set). Behaviour unchanged.
B4	2026-11-05	LIC01 manual→auto: the active setpoint (`Input.Setpoint`) starts at the PV and ramps to the operator's `Command.Setpoint` (`Config.SetpointRamp`); the PLC no longer overwrites the operator's setpoint. Pump count status (next stop / start / rotation, time left, why paused) in `Pen.InletSupervisor` and per pump in `InletPump.Status`. Pump Start / Stop written in auto are cleared. `InletPumpEnable` defaults to TRUE.
B5	2026-11-05	Pen level limits (configurable): WL 0.03, AL 0.02, WH 0.08, AH 0.09, AHH 0.10 m (ALL 0.01 m, close the outlet, not implemented); warnings clear by themselves, alarms are acknowledged; AHH trips the inlet pumps and stops the pen (I8). The level setpoint is kept within WL..WH.
B6	2026-11-05	Framo has no HOLD any more ([FRAMO-FD] rev. 04, [FRAMO-SCD] A03): on heartbeat loss the PCC keeps the pump running at its last setpoint × 0.8

2 Block diagram description

2.1 The process in brief

The pen PEN01 is a closed bag of about 20 000 m^3 (Ø 50.5 m, depth 17.3 m) with a Ø 5.5 m outlet in the bottom. Sea water is pumped in from three separate intakes at 10–30 m depth. Each intake has a fish screen (XF-3n), a tunnel to a Framo inlet pump (PM-3n, rated 20 000 m^3/h at 130 rpm, 60 kW rim motor) and a riser with a gooseneck over the floating collar into the pen.

The water inside the pen is kept a few centimetres higher than the sea outside. This *over-height* presses the bag outwards so that it keeps its shape, and it drives the outflow through the bottom outlet. The outlet dampers stand at a fixed opening for each operating mode; the level is therefore controlled with the *pump speed*: more speed gives more inflow, a higher level and more outflow, until inflow and outflow balance [FD] §3.1.

Before a pump can lift water over the collar, the air in the riser must be removed. Each inlet has a vacuum (air evacuation) system run by the Framo controller: a vacuum chamber TV-3n on top of the riser, an ejector EJ-3n driven by compressed air through valve XV3nA, level switches LSL3n/LSH3n and a vent valve XV3nB that breaks the vacuum when the pump stops, so the pen does not drain back through the intake [FD] §3.2.1.1.

The operating modes set the flow through the pen and the level setpoint (Table 2).

Table 2: Operating modes [FD] Tabell 10, with the operating point of the design basis [DES].

Mode	Res. time [min]	Flow [m^3/h]	SP ΔH [cm]	Pumps [–]	Speed n_0 [rpm]	Damper [%]	Comment
O-35	35	34 000	6.0	3	73.7	100	flow above the [DES] maximum[OI12]
O-45	45	27 000	2.5	2	87.8	100	
O-60	60	20 000	0.5	2	65.0	91	5 mm setpoint[OI7]
S-75	75	16 000	0.5	2	52.0	73	service mode[OI7]

With two pumps running, [FD] asks for the pumps to be alternated so that the third does not stand still for too long (HOLD in [FD])[OI15].

2.2 Physical units and networks

Figure 1 shows the physical units and how they are connected; Table 3 gives the role of each.

2.3 Simplified process and instrument diagram

Figure 2 is a simplified P&ID for one inlet ($n = 1$); inlets 2 and 3 are identical. Tags follow ISA-5.1 as on the P&ID set [PID]. Bubbles: plain = field instrument, with a bar = function in the Heimdall PLC, grey = function in the Framo controller.

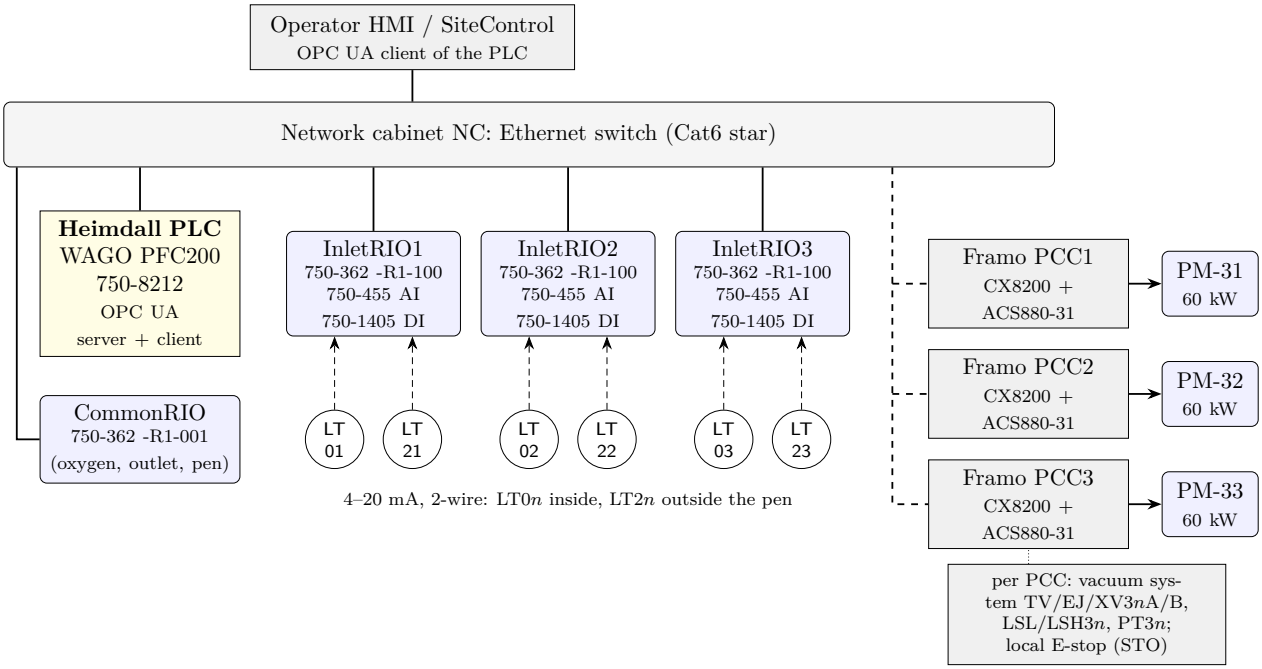


Figure 1: Physical units and networks of the inlet system (after [PID] CAS-LAY-700 and [EL]). Solid: Ethernet / Modbus TCP (PLC ↔ RIO, 100 ms); dashed: OPC UA (PLC ↔ Framo PCC); dotted arrows: 4–20 mA. IP addresses are not on any drawing[OI23].

Table 3: Physical units of the inlet system.

Unit	Role
Framo PCC1–3	One cabinet per inlet [EL] 874.001: Beckhoff CX8200 PLC (TwinCAT 3, OPC UA server) and ABB ACS880-31-105A-3 drive for the rim motor of PM-3n, local emergency stop pushbutton with Safe Torque Off (XSTO). Runs the pump's own start and stop sequences, the vacuum system and the protections, and publishes the pump state. Framo scope.[OI19]
Framo inlet pumps PM-31/32/33	SR2000 rim-driven propeller pumps, 40–130 rpm, 20 000 m^3/h at 130 rpm, 72-pole motor (78 Hz at 130 rpm) [DES].
Vacuum system (per inlet)	TV-3n, EJ-3n, XV3nA (air to ejector), XV3nB (vent), LSL3n, LSH3n, PT3n (compressed air). Controlled by the PCC; the PLC only reads status.[OI11]
Heimdall PLC	WAGO PFC200 750-8212 in the PLC/RIO1 cabinet (UPS backed). Runs the level loop, the pen sequence and the pump supervision. No local I/O.
InletRIO1–3	750-362 Modbus TCP couplers, one per inlet. 750-455 analog inputs for LT0n (word 0) and LT2n (word 1); 750-1405 for cabinet power status (word 4) and bilge switches (word 5). Polled by the PLC every 100 ms.
CommonRIO	750-362 coupler for the common signals (oxygen, outlet, analysis). Not used by the inlet functions.
Level transmitters	LT01–03 inside the pen (VEGAPULS radar on the gooseneck), LT21–23 outside (submersible pressure probe), one pair per inlet, 0–2 m, 4–20 mA 2-wire.[OI18]
HMI / SiteControl	Operator interface; reads and writes the PLC's variables over OPC UA.

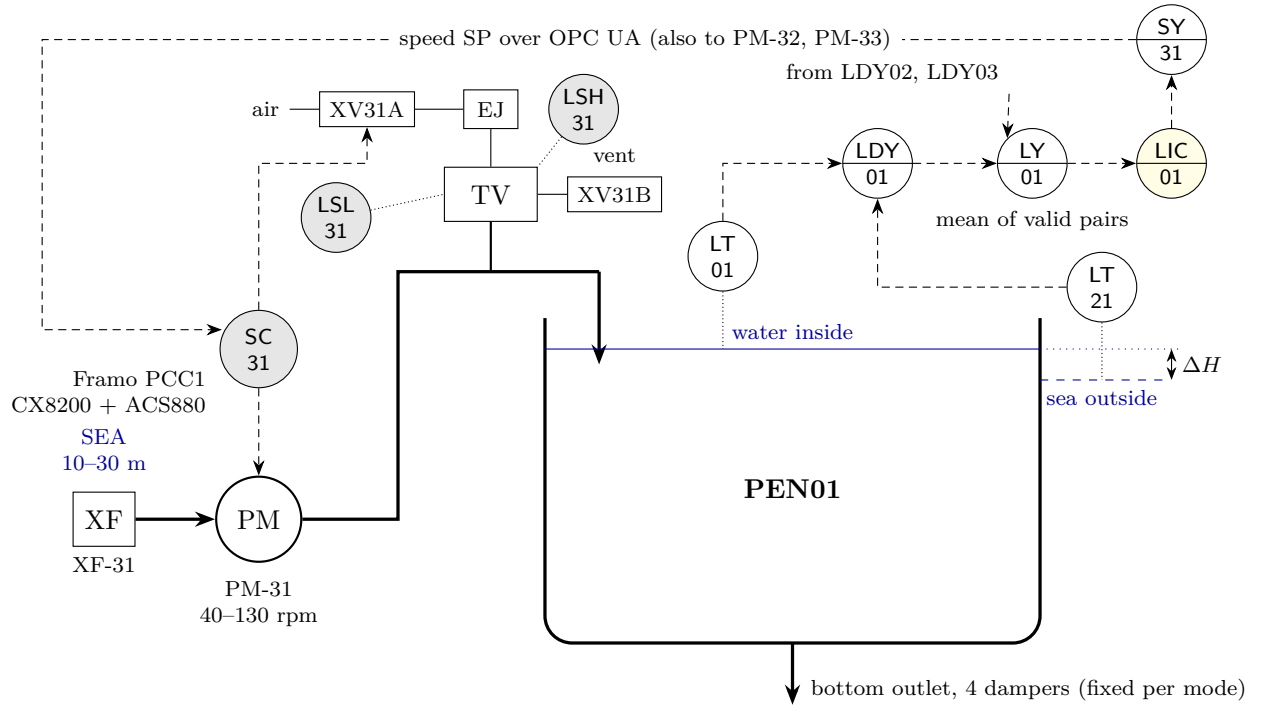


Figure 2: Simplified process and instrument diagram, inlet 1 ([PID] CAS-PID-100 and -400). LDY0n forms the over-height of pair n, LY01 the mean of the valid pairs, LIC01 the total speed, shared as the common speed setpoint SY3n of the pumps in REMOTE.

3 Functional diagram description

3.1 Top-level functions

Figure 3 shows the functions of the inlet system and the data between them. The functions F1–F3 run once per pen; F4 and F5 once per pump. The Framo controller (grey) is outside the Heimdall PLC.

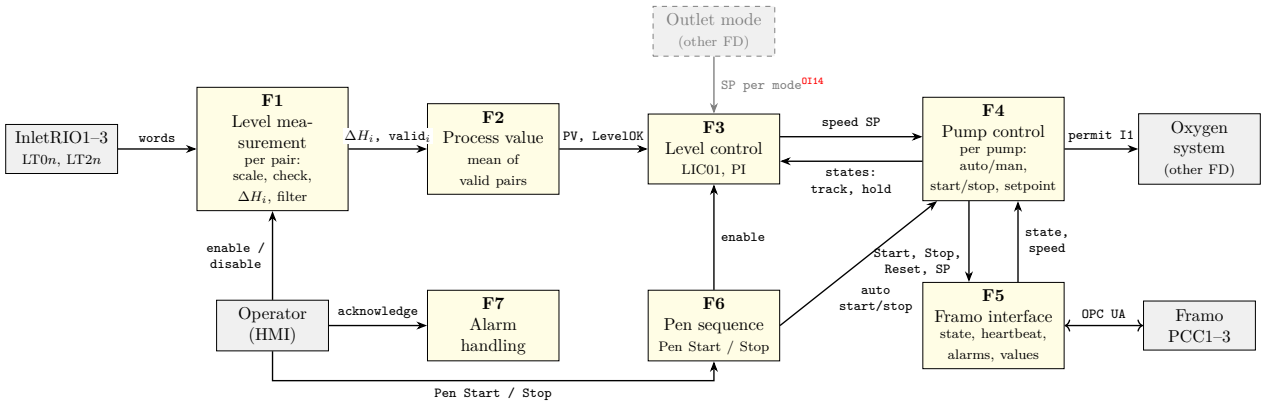


Figure 3: Top-level functional block diagram of the inlet system ([COO] Fig. 7.8 style). Not drawn: alarm conditions from F1–F5 to F7, and the operator's setpoint, LIC01 auto/manual and pump commands to F3 and F4.

Table 4: Functions and where they are implemented in [PLC].

	Function	What it does	PLC code unit
F1	Level measurement	Scales LT0i and LT2i, checks their health, forms ΔH_i and filters it.	`FB.Inlet.InletSensors`, `AnalogSensor`
F2	Process value	Mean of the valid ΔH_i; LevelControlOK.	`FB.Pen.PenControl`
F3	Level control	Which pumps count and which the PLC sets; LIC01 (total speed) with its modes; share per pump in REMOTE; operator auto/manual; deviation alarms.	`FB.Inlet.InletPumpSet`, `FB.Inlet.PenLevelControl`, `ControllerMode`
F4	Pump control	Auto/manual, start/stop latch, speed setpoint per state, start/stop timeouts, oxygen permit.	`InletControl.InletPumpControl`
F5	Framo interface	Writes commands, reads state, alarms and values; heartbeat supervision.	`PRG.FramoDataProcessing`, `InletControl.Heartbeat`
F6	Pen sequence	Pen state machine (Start/Stop, Oxygen Start/Stop), auto start/stop of the pumps in auto, automatic pump count and rotation.	`FB.Inlet.PenSequence`, `FB.Inlet.PumpCount`; order: `FB.Inlet.InletSupervisor`
F7	Alarm handling	Alarm state and acknowledgement per alarm.	`AlarmManagement` instances

3.2 Level control loop LIC01

Principle. One PI controller, LIC01, holds the over-height at the setpoint. Two rules cover every pump state (DEC5, DEC9):

- **R1, sharing.** A pump *counts* when it is in auto and runs: START, REMOTE or STOP, or its link is lost while it counted (the PCC then runs it on its own: in LINK LOST at the estimated speed, last setpoint × `CommLossFactor`, otherwise its last measured speed). LIC01's output T is the *total* speed of the counted pumps (sum of rpm). The counted pumps in REMOTE with a live link are *controlled* (set R); they share what is left after the others' speed, so all of them run at the same speed [FD] §3.2.2:

$$n_{\mathrm{SP}} = \mathrm{limit}_{40}^{130}\Big(\frac{T - \sum_{j \notin R} n_j}{|R|}\Big)$$

- **R2, mode.** LIC01 controls only while a counted pump is in REMOTE; otherwise it tracks the summed speed of the counted pumps.

Inflow goes with the total, so the loop gain does not depend on the number of pumps (one parameter set instead of [FD] Tabell 8). A pump that changes group leaves T as it is: a pump that trips leaves the sum and the others take over its share in the same cycle; a starting pump adds its speed while it ramps up and the others give way; a pump that loses its link stays in the sum at 0.8 of its last setpoint and the others correct for it; when its link is back it is in REMOTE again, takes its share and the speeds even out with no step in the total. The three PCCs do not know each other's state, so the PLC keeps controlling with the pumps it can set (DEC9, 2026-10-23; this replaces the freeze of [FD] Tabell 5). The PLC reads the pump states directly (`InletPumpSet`), so a pump counts from the cycle its state changes. Figure 4 is the control scheme.

Process value. At each inlet i the PLC computes the over-height of the pair from the scaled, unfiltered readings and filters it with a 2nd-order Butterworth low-pass (cut-off $f_c = 0.05$ Hz):

$$\Delta H_i = \mathrm{LT0}i - \mathrm{LT2}i, \qquad \mathrm{PV} = \Delta H_{\mathrm{avg}} = \frac{1}{|V|} \sum_{i \in V} F\{\Delta H_i\}$$

where V is the set of *valid* pairs. A pair is valid when both transmitters are enabled and healthy (750-455 status bits 1..0 = 00) and the pair itself is enabled. If either transmitter of a pair is lost, the whole pair drops out of the mean[DEC1]. [FD] asks to filter each transmitter before the difference; since the filter is linear, filtering the difference gives the same result while both are healthy, and it restarts cleanly when a pair returns [DES] §7.

Setpoint. The over-height setpoint in m (`Pen.LevelController.Command.Setpoint`, retained, default 0.06 m = O-35). [FD] wants it selected from the operating mode, with a manual override; today the operator enters it[OI14].

Output and modes. Total speed, limited to $\sum_{j \notin R} n_j + |R| \times$(40–130) rpm. The per-pump values (limits, `SafeOutput`, the operator's manual output) are rpm per controlled pump. LIC01 has one mode at a time (`LevelControlMode`), by priority:

Mode	When	Output T
Idle	pen stopped, or no pump counted	0
Manual	operator manual	$\sum_{j \notin R} n_j + \lvert R \rvert \times$ `ManualOutput`
SafeOutput	no valid level pair[DEC3]	$\sum_{j \notin R} n_j + \lvert R \rvert \times$ `SafeOutput` (60 rpm)
Tracking	pumps counted, none controlled (start-up, all links lost)	summed speed of the counted pumps
Auto	otherwise	PI

Every return to Auto is bumpless: `PIDController` restarts the PID from the output it had.

Operator auto/manual, bumpless both ways (`ControllerMode`; while the pen is started and a pump is controlled):

- Auto→manual: the manual output is set to the current share, so the pumps keep their speed.
- Manual→auto: LIC01 restarts from the manual output with a cleared integral. Its active setpoint (`Pen.LevelController.Input.Setpoint`) starts at the current over-height (PV), if a level pair is valid, and ramps to the operator's setpoint (`.Command.Setpoint`) at `.Config.SetpointRamp` (0.01 m/min; 0 = at once). It starts with no error and no step, and the operator's setpoint is kept: the PLC never writes `Command.Setpoint`.

With the pen stopped a mode change leaves the setpoint and the manual output as they are.

Level deviation (DEC8). With LIC01 in Auto, alarm *Level deviation high* (*low*) when ΔH_{avg} stays more than `LevelDeviationLimit` (0.02 m) above (below) the setpoint for `LevelDeviationDelay` (5 min): the pumps cannot hold the level, at maximum speed or at minimum speed with no pump left to stop. Only in Auto.

Level limits (`Pen.Config`, on ΔH_{avg}, with a valid measurement, each for `LevelAlarmDelay` 10 s, on the over-height filtered at 0.05 Hz per pair, so waves do not reach them; an active limit clears only `LevelAlarmHysteresis` 0.005 m back inside it). Warnings clear by themselves, alarms are acknowledged: WL 0.03 m, AL 0.02 m (ALL 0.01 m, close the outlet valve, not implemented), WH 0.08 m, AH 0.09 m and AHH 0.10 m. AHH trips: all inlet pumps stop, also in manual, and the pen stops (`Pen.Status.LevelHighTrip`); the pen cannot be started again until the level is below WH. The level setpoint is kept within WL..WH: the PLC takes a value outside back to the limit.

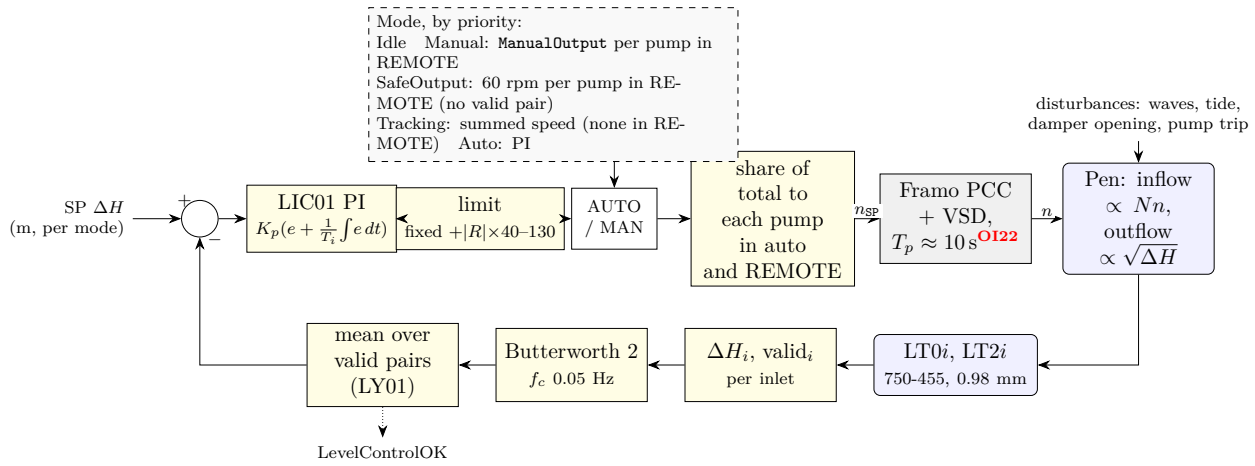


Figure 4: Control scheme of LIC01 ([PDS] SMs 1–10, [DES] §7). R = pumps in REMOTE; fixed = speed of the other counted pumps.

3.3 Pen start and stop sequence

Figure 5 shows the sequence that the operator's Pen Start and Pen Stop commands run. Pumps in manual are left out of it; the operator starts and stops them one by one.

Pen states ([COO] §7.3.7; `Pen.Status.State`, `FB.Inlet.PenSequence`): *Stopped* → *Starting* on Start → *Running* when a counted pump is in REMOTE (LIC01 takes over) → *Stopping* on Stop → *Stopped* when no pump in auto runs. The state machine runs last in `InletSupervisor`; level control, pump count and oxygen act on the new state from the next cycle. Starting falls back to Stopped if no pump starts at all. Start is accepted only in Stopped and when `Pen.Status.Ready`: an enabled pump in auto is AVAILABLE (`PumpReady`) and a level pair is valid. Oxygenation is switched for the whole pen with Oxygen Start / Stop (accepted while Starting or Running; Stop turns it off).

Command rules. Every operator command is a pulse that the PLC clears in the same scan; one it refuses (wrong state, not ready) sets `Pen.Status.CommandRejected`. Modes are statuses: LIC01, OIC01, the pumps, the oxygen valves and the outlet go to manual and back with `Command.Manual` / `Command.Auto` pulses and show `Status.Manual`. Enable / disable of a pump or oxygen line is retained configuration (`Inlet[i].InletPump.Config.InletPumpEnable`, `OxySystem[i].Config.Enable`). Commands the PLC code gives itself are in `InternalCommand` (supervisor → pump: `AutoStart` / `AutoStop`; pump → PCC: Start, Stop, Reset, speed, heartbeat), never in `Command`.

Fresh site (configuration set, pumps stopped; after a download pumps, lines and controllers are in auto): acknowledge the start-up alarms, check *Ready*, Pen Start, then Oxygen Start.

3.4 Pump count and rotation

While the pen runs, the PLC chooses how many pumps run and which (DEC7). The goal is to run the pumps near their operating speed rather than many pumps slowly. The PLC acts only on pumps that are *candidates*: in automatic, not UNAVAILABLE or LOCKDOWN, and enabled for it with the retained flag `Inlet[i].InletPump.Config.InletPumpEnable`. Pumps in auto without Enable follow Pen Start / Pen Stop only; pumps in manual are left alone. The logic is in `FB.Inlet.PumpCount`.

- **Pen Start** starts every candidate (start all, then reduce). A pump that becomes a candidate while the pen runs (back to auto, reset from LOCKDOWN, enabled) is started too.
- The count logic acts only with LIC01 in auto and every pump where it is wanted (in REMOTE

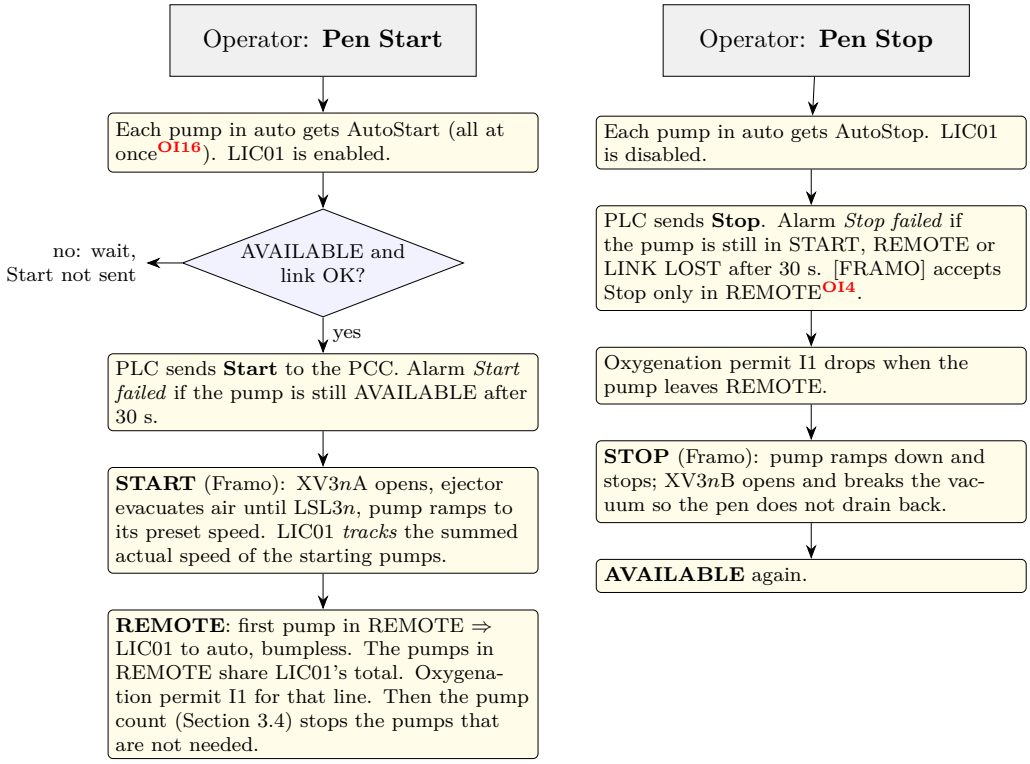


Figure 5: Pen start (left) and stop (right) sequence [FD] §3.2.1, [PLC] InletSupervisor/InletControl.

or AVAILABLE, none starting or stopping).

- **Stop a pump** when the others would run at or below `PumpTargetSpeed`: $T/(N-1) \leq 85$ rpm for `PumpRemoveDelay` (15 min). The pump with the most running days (`MotorRunningDays`) stops.
- **Start a pump** when the share is at or above `PumpAddSpeed`: $n_{SP} \geq 115$ rpm for `PumpAddDelay` (2 min), and a candidate is AVAILABLE. The one with the fewest running days starts. Starting is the safe direction and therefore faster than stopping.
- `PumpAddSpeed` above `PumpTargetSpeed` gives the hysteresis: 2 pumps at 115 rpm → 3 at 77 rpm; 3 at 50 rpm → 2 at 75 rpm; neither falls back at once.
- **Rotation.** When a running candidate has been in REMOTE for `PumpCycleTime` (24 h) and a candidate is AVAILABLE, the standby with the fewest running days starts first; the pump that has run its cycle stops once the standby is in REMOTE (make before break). The sharing keeps the total speed through the change. If the standby does not start, the running pump carries on.
- **Start failed.** A candidate still AVAILABLE `StartupTime` + 10 s after its start is left out of the count (alarm *Start failed*); the logic starts another one if the share stays high. It is taken back when it is no longer a candidate (operator: manual, disable) or the pen stops.

The HMI shows the number of pumps the PLC wants (`Pen.InletSupervisor.PumpsWanted`) and a rotation in progress (`PumpRotationActive`). What the count will do next is status in `Pen.InletSupervisor` (pump numbers 1–3, 0 = none): `CountActive` and why not (`CountBlocked`: pen not started, LIC01 not in auto, pumps changing, rotation running); the stop pending (`StopPending`, time left `StopIn`, the pump `StopNext`, and `StopCondition` = Total / (pumps − 1) against `PumpTargetSpeed`); the start pending (`StartPending`, `StartIn`, `StartNext`, `StartCondition` = the share against `PumpAddSpeed`); the next rotation (`RotationIn`, `RotationOut`, `RotationStandby`). Per pump in `Inlet[i].InletPump.Status`: `AutoWanted`, `AutoStartFailed`, `AutoStartTimeLeft` and `TimeInRemote`. [FD]'s warning for a pump idle too long and the number of pumps per operating mode are not implemented[OI15].

3.5 Framo pump state machine

The Framo controller owns the pump state and publishes it as one Boolean per state; the PLC turns the flags into `Inlet[i].InletPump.State` and asks for transitions with Start, Stop and Reset. Figure 6 is the state diagram of [FRAMO] rev. B without HOLD, which [FRAMO-FD] rev. 04 removes; red dashed labels mark where [FD] or the PLC expect something else. LINK LOST (4) is the PLC's own state for a pump that runs without a link. Table 5 gives what the PLC does in each state.

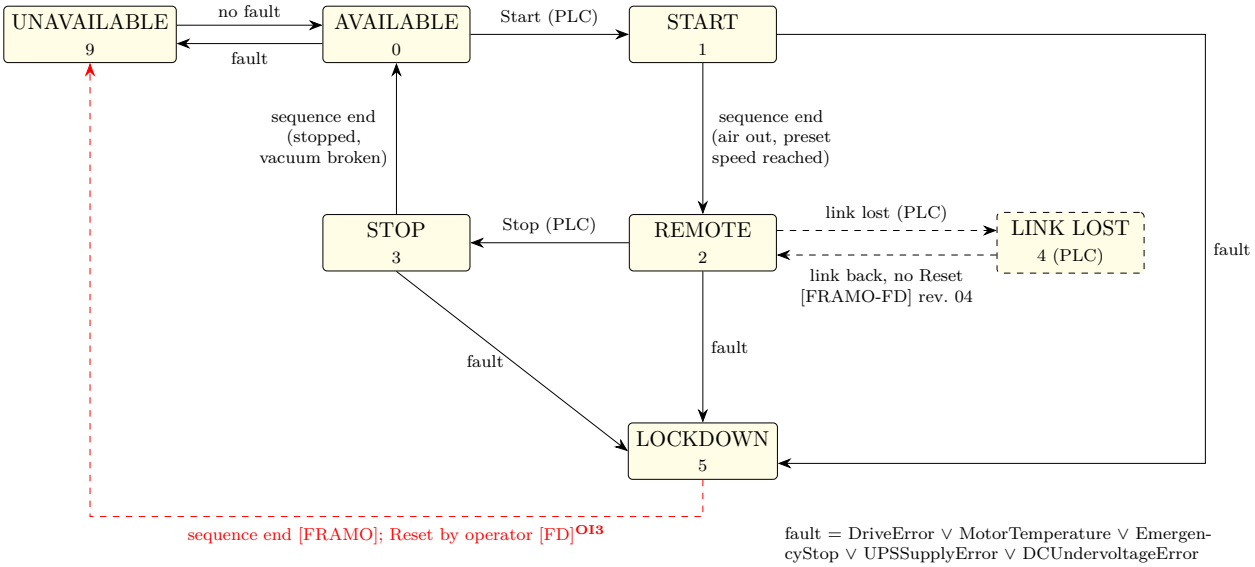


Figure 6: State transition diagram of the Framo inlet pump [FRAMO] rev. B p. 6, [COO] §7.2.3. Numbers are the values of `ENUM.InletPumpState`. Without a heartbeat answer from that PCC the PLC shows a pump that was starting or running as LINK LOST (the PCC keeps it running at 0.8 of its last setpoint) and any other pump UNAVAILABLE, whatever the PCC publishes.

3.6 Heartbeat

The PLC and each PCC supervise the link between them with a heartbeat. As implemented in the PLC (2026-10-15), the heartbeat is *one shared Boolean* `Framo_n.HeartbeatToggle` (Figure 7):

- the PLC sets it when it has been FALSE for `HeartbeatInterval` (1 s);
- the PCC clears it when it is TRUE;
- the bit always says whose turn it is, so after a (re)connect neither side waits for the other;
- only the PCC's clear (falling edge) proves that the PCC is alive. When the PCC has not cleared the bit for `HeartbeatTimeout` (10 s), the PLC raises *Communication loss*, shows a running pump LINK LOST (any other UNAVAILABLE), sends no Start, keeps the pump in LIC01's total at last setpoint × `CommLossFactor` (DEC9) and removes the oxygenation permit of the line;
- the PCC, seeing no set from the PLC within its timeout, keeps the pump in normal operation at its last setpoint × 0.8, or at the last setpoint if CAS has set *stand-alone* ([FRAMO-FD] rev. 04 §7–8, [FRAMO-SCD] sheet 3). When the heartbeat is back the PCC follows the setpoint again and resets stand-alone; no Reset is needed. The stand-alone command has no node yet, so the PLC does not send it[OI1].

The timeout is 10 s instead of the 5 s DUT default because the round trip through the CODESYS Data Sources takes about 4.6 s in the test rig. [FRAMO] rev. B lists `HeartbeatToggle` (BOOL, RW), `HeartbeatInterval` and `HeartbeatElapsed`; [FD] describes an INT counter 0–32767. The

Table 5: PLC action per pump state ([PLC] `InletControl.InletPumpControl`).

State	Speed setpoint sent	Start	In LIC01	Notes
UNAVAILABLE	0	no	no	Also shown while the link (heartbeat) is down, for a pump that was not running.
AVAILABLE	actual speed	yes	no	Setpoint follows the actual speed so the take-over is bumpless [FD] Tabell 5.
START	actual speed	yes	yes	LIC01 tracks the summed speed of the starting pumps; its speed is subtracted from the total.
REMOTE	share of LIC01's total (auto) or operator setpoint (manual)	yes	yes	The only state in which the PLC sets the speed. Oxygenation permit I1 with a live link.
LINK LOST	last setpoint (sent again when the link is back)	no	yes	Heartbeat lost while starting or running. The PCC runs the pump at last setpoint × 0.8 ([FRAMO-FD] rev. 04; in stand-alone the last setpoint); the PLC counts it in LIC01's total at that estimate (`ActualSpeed`), the pumps in REMOTE correct for it (DEC9); oxygenation stops; no Reset needed.
STOP	0	no	yes	
LOCKDOWN	0	no	no	Needs Reset after the fault is cleared[OI3]. The only state in which the PLC sends Reset.

baseline must be agreed with Framo[OI1].

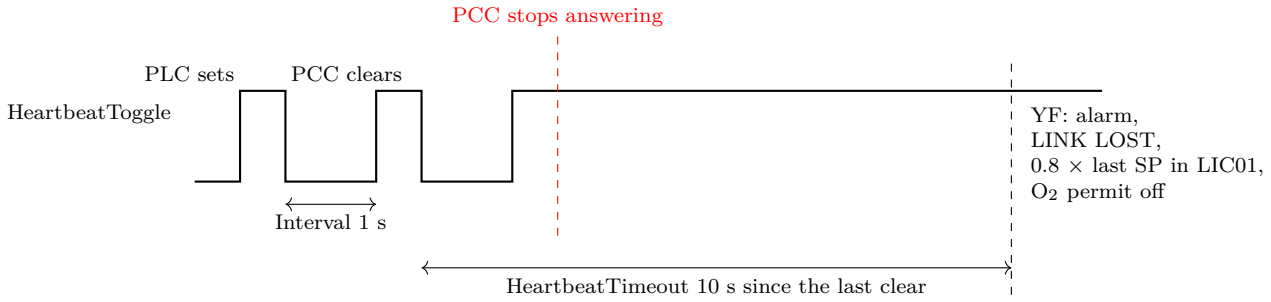


Figure 7: Heartbeat as implemented in [PLC]: one shared bit, the PLC sets it, the PCC clears it.

3.7 Planned: pump state from the SCD signals

Status: planned, not implemented. [FRAMO-FD] rev. 04 and [FRAMO-SCD] A03 are taken as the description of the Framo pump control system. They have no pump states any more (no UNAVAILABLE, AVAILABLE, START, STOP, REMOTE, HOLD or LOCKDOWN); LOCKDOWN remains only as Framo's reaction to a system-wide failure. The PLC therefore works out the pump state itself from the signals of the SCD. Until Framo issues the OPC UA interface list for rev. 04, the PLC keeps reading the state flags of [FRAMO] rev. B (Section 3.5) and the simulator keeps that node set.

The SCD in brief.

- **Start** (sheet 2): CAS Start (PA003-M01-U04) runs the air evacuation (ejector air valve UY31

open, release valve UY32 closed) until the vacuum chamber's high switch LSH32 shows water; then the drive gets its run command. *Starting sequence active* (UI010) while this runs; *failed to start* (UA011, action alarm) if priming takes too long.
- **Stop** (sheet 2): CAS Stop (U05) stops the drive; when the pump no longer runs, UY32 opens and breaks the vacuum. *Sensor deviation* (UA001A) when the chamber switches LSL31 and LSH32 disagree.
- **Setpoint** (sheet 3): the drive gets the CAS setpoint (U56) only with CAS *Remote control* (UY012) and *ready to remote control* (UI013: running above 40 rpm for 30 s); otherwise the start setpoint (40 rpm, set at commissioning). Heartbeat fault: last setpoint × 0.8; with CAS *Stand-alone* (UY014): the last setpoint, reset when the heartbeat is back.
- **Commands from CAS**: Start, Stop, Reset (a 1 s reset pulse to the drive on the rising edge), Inhibit start (M01-Q03, no start possible, also not locally; a running pump is not stopped), Remote control, Stand-alone, speed setpoint, heartbeat.
- **Trips** (sheet 4, action alarms): earth leakage, short circuit, safe torque off, motor stall, power unit loss, overspeed, input / output phase loss, drive communication fault. Earth leakage also as a warning. Drive feedback: ready (Q65), running (Q02), tripped (Q10), in remote (Q27), start inhibit (U03).

Planned PLC pump state (Table 6). The state names used by LIC01, the pump count and the oxygen permit stay; LOCKDOWN becomes *Fault* and a *Local* state is added.

Table 6: Planned PLC pump state from the SCD signals ([FRAMO-SCD] A03; signals to be confirmed on the interface).

State	From the SCD signals
Unavailable	stopped and drive not ready, not in remote, tripped or start-inhibited; or no link and not running
Available	drive ready, in remote, not tripped, no start inhibit, not running, no starting sequence (or one ready-for-start status from Framo, asked for)
Starting	starting sequence active (priming), or running but not yet ready to remote control (40 rpm phase); LIC01 tracks
Remote	running with the remote setpoint in use: the PLC sets the speed (LIC01 share or manual)
Local	running on the start / local setpoint without Remote (e.g. Remote dropped); counted at its actual speed
Stopping	Stop sent and the pump still running (spin-down), until *pump running* drops
Link lost	heartbeat lost while starting or running: last setpoint × 0.8, or the last setpoint in stand-alone (as now)
Fault	drive tripped or failed to start (replaces LOCKDOWN); leaves with Reset or a new Start

What the SCD asks of the PLC.

- Write Remote control TRUE continuously while CAS controls the pump; after a reconnect or PLC restart, control returns without a new command if Remote is still TRUE and the pump is ready (to be confirmed by Framo).
- Expect at least 30 s at 40 rpm after each start before the pump takes the CAS setpoint; LIC01 tracks meanwhile.
- After a PLC restart, write a setpoint that starts from the pump's actual speed, so the first cycles with Remote and ready do not step the pump.
- Drive Inhibit start from the start interlocks: pen level AHH trip, pump out of service; outlet closed and pen stopped to be decided (they also block a local start for service).
- Send Reset after a drive trip; decide when to use Stand-alone.
- Keep the history of the pumps: [FRAMO-FD] leaves historical logging to CAS.

Open with Framo: the interface list, a ready-for-start status, the drive feedback and alarms on the

interface, Remote level or edge, the start setpoint as a CAS parameter, and the status of emergency stop, lockdown and drive trip (questions of 2026-11-05).

3.8 Operating modes

The operating modes (Table 2) are selected on the outlet function. A mode sets the damper opening and, per [FD], the level setpoint of LIC01 and the number of pumps (2 or 3). In [PLC] the outlet has the mode table with damper positions; the level setpoint is not taken from the mode yet[OI14]. The number of pumps follows from the flow needed (Section 3.4) instead of from the mode.

4 Functional requirements

4.1 Processes

P1 The system shall pump sea water into the pen with up to three Framo inlet pumps; normally 2 or 3 run, depending on the operating mode.

P2 The system shall hold the over-height ΔH_{avg} at the setpoint by setting one common speed for all pumps in REMOTE in level control, their share of LIC01's total speed.

P3 The system shall start and stop the pumps in automatic from one Pen Start / Pen Stop command, and let the operator start, stop and set the speed of a pump in manual.

P4 While the pen runs, the system shall run only as many pumps as the flow needs, near their operating speed, and rotate the running pumps by run time (Section 3.4).

P5 The system shall supervise the link to each Framo controller and react to its loss (LINK LOST).

P6 The system shall give each oxygen line its permit only while its inlet pump runs in REMOTE with a live link (interlock I1).

P7 The system shall present the pump state, the Framo values and alarms, and the levels on the HMI.

4.2 Inputs

4.3 Outputs

- Per pump, to PCCn over OPC UA: Start, Stop, Reset, SpeedSetpoint [rpm], HeartbeatToggle (set).
- To the oxygen system: `Inlet[i].InletPump.Status.OxyPermit` (permit I1).
- To the HMI: states, PV, setpoint, LIC01 output and mode, alarms.

The inlet system has no hard-wired outputs; the PLC has no command path to the pumps other than OPC UA. A lost link therefore always leaves the pump to the PCC (LINK LOST: last setpoint *imes* 0.8).

4.4 Startup and shutdown

PLC start (power-up, download). Configuration and the level setpoint are retained. The pen start latch is not retained: after a PLC restart no pump is started automatically. [FD] Tabell 2–3 asks for auto-restart after a power failure, in a sequence to avoid simultaneous inrush currents; this

Table 7: Inputs to the inlet functions (details in Tables 13 and 14).

Input	Source	Used for
LT01–03, LT21–23	InletRIO *n* words 0, 1	PV of LIC01, display, alarms
Pump state flags (6 used per pump; HOLD no longer read)	PCC*n*, OPC UA	F4, tracking, start permissive
MotorVelocity	PCC*n*, OPC UA	tracking, bumpless transfer, display
Framo alarms (11 per pump)	PCC*n*, OPC UA	alarms, info panel
HeartbeatToggle	PCC*n*, OPC UA	link supervision
Vacuum status (LSL, LSH, V001, V002, air pressure)	PCC*n*, OPC UA	display; overfill alarm[OI11]
Drive values (current, power, ...)	PCC*n*, OPC UA	info panel
24 V field breaker XF05 status	InletRIO *n* word 4.0	alarm (it feeds the LT loops)
Operator commands and parameters	HMI, OPC UA	Table 15

is not implemented[OI16]. The PLC shows every pump UNAVAILABLE until its PCC has answered the heartbeat once, so no Start is sent on stale data.

Normal start and **normal stop**: Figure 5.

Pump start while others run. A pump started (in auto or manual) while another pump is already in REMOTE does not interrupt automatic control: while it starts its speed counts in LIC01's total and the pumps in REMOTE give way; it reaches REMOTE at about its share.

Emergency stop. Each PCC has a local E-stop pushbutton acting on the drive's Safe Torque Off; [EL] 874.001 notes that multiple E-stop actuators are connected in series. Whether the E-stops stop all three lines, and how the oxygen line is stopped on E-stop through the Heimdall PLC, is to be confirmed[OI13]. The PCC reports `EmergencyStop`; the pump goes to LOCKDOWN (or UNAVAILABLE if it was not running).

4.5 Fault identification and recovery

Table 8: Faults, detection, automatic response and recovery.

Fault	Detection	Automatic response	Recovery
One level transmitter bad	750-455 status bits ≠ 00 (wire break, over-range) for 1 s	Quality alarm; its *pair* drops out of the PV; LIC01 continues on the remaining pairs	Automatic when healthy; the filter restarts at the current value (no dip)
Transmitter disabled by operator	Config.Enable FALSE	Pair drops out	Operator enables it
All pairs lost	no valid pair	Alarm *Level measurement lost* (LAL01); LIC01 to manual at `SafeOutput` 60 rpm (interlock I6)	Back to auto bumplessly when a pair is valid

Fault	Detection	Automatic response	Recovery
InletRIO n link lost	Master not connected, or no new response for 2 s while the master still reports connected (freshness watchdog on the master's receive counter, `HasReceivedChannelData`, `Status.DataStale`)	Alarm *InletRIO comm. error*; both transmitters of inlet n count as bad, so its pair drops out of the PV (DEC6); the master is restarted at once (disabled 1 s, channels added again) and every 10 s while stale (`Status.Restarts`). The station's RIO and oxy pump drive have one master each and restart independently: a restart of one does not interrupt the other	Automatic
24 V field breaker XF05 tripped	InletRIO word 4.0	Alarm; both transmitters of that inlet read bad ⇒ pair drops out	Reset XF05 (HMI, DO 0.0)
Framo link / heartbeat lost	no clear for 10 s, or the data source `Framo_n` not connected	Alarm *Communication loss*; PLC shows a running pump LINK LOST; it stays in LIC01's total at last setpoint × 0.8 and the pumps in REMOTE keep controlling (DEC9); with no pump left in REMOTE LIC01 tracks; oxygenation permit off. PCC: normal operation at last setpoint × 0.8 (stand-alone: last setpoint)	Link back: the pump is in REMOTE, takes its share, the total is unchanged; no Reset
Pump trip / fault	PCC alarm flags; state LOCKDOWN	Common alarm *Pump n failed*; pump leaves level control; the remaining pumps take over its share in the same cycle (DEC5); if their share stays ≥ `PumpAddSpeed`, a standby candidate is started; permit off	Fix the fault, operator Reset[OI3]; as a candidate it is started again while the pen runs
Start not acknowledged	Start sent, still AVAILABLE after 30 s	Alarm *Start failed*; a candidate is left out of the pump count and another one is started if the share stays high (Section 3.4)	Operator
Stop not acknowledged	Stop sent, still in START, REMOTE or LINK LOST after 30 s	Alarm *Stop failed*	Operator, locally at the PCC
Reset not answered	Reset given in LOCKDOWN, pump still in that state after 2×`ResetExpectedTime` (10 s)	Reset withdrawn; warning *Failed to reset* (no acknowledge)	Check the PCC; give Reset again
Speed deviation	$\lvert n_{SP} - n \rvert >$ `SpeedDeviationLimit` (10 rpm) for `SpeedDeviationDelay` (30 s), in REMOTE	Alarm *Speed deviation*	Operator; automatic when the pump follows again
Vacuum chamber overfill	LSH3n	not implemented (5 s delay in [FD])[OI11]	–

Fault	Detection	Automatic response	Recovery
Emergency stop	PCC `EmergencyStop`	Pump stops (STO); alarm; LOCKDOWN	Release E-stop, Reset[OI13]
LIC01 at its limit	`LimitsActive`	Alarm; integral held (anti-windup of `Util.PID`)	Automatic
Level not held	$\lvert\Delta H_{\text{avg}} - \text{SP}\rvert >$ the limit for the delay, LIC01 in auto	Alarm *Level deviation high / low*	Operator; automatic when back within the limit

4.6 Interlocks

ID	Interlock	Source	In PLC
I1	Oxygenation permit line n = PM-3n in REMOTE AND heartbeat OK.	[FD] §3.2.1, [PID] -100	yes
I2	2oo3 high-high LS7nmHH in inlet box n stops PM-3n.	[FD] §3.2.6, [PID] -600	no[OI20]
I3	Group start only with the outlet open.	PLC function flow diagram	no[OI20]
I6	No valid level pair: LIC01 to manual at SafeOutput.	[PID] -400	yes
I8	Over-height above LevelAHH: all inlet pumps stop, the pen stops, until below LevelWH.	PLC (2026-11-05)	yes
–	Start only when the pump is AVAILABLE and the link is OK.	[FD] Tabell 5	yes
–	Emergency stop: Framo E-stop stops the pump (STO); oxygen line to stop.	[FD] Tabell 2, [EL]	partly[OI13]

4.7 Communication

- **PLC ↔ Framo PCCn:** OPC UA. The PLC is the client (CODESYS Data Sources `Framo_1..3`); each PCC is a server (TwinCAT TF6100). Signal list: Table 14. The data sources use a dynamic configuration filled by `FramoLink` (in `FramoDataProcessing`): endpoint `opc.tcp://<FramoIP>:<F` and the subscription's sampling and publishing interval `FramoUpdateRate`, all from the retained `Inlet[i].InletPump.Config`, so the PCC address can be changed from the HMI without a download. A new endpoint or update rate, or a link lost for 30 s (and every 30 s after), disconnects the data source for 1 s and connects it again (`InletPump.Status.FramoRestarts`). The link counts as live only with the heartbeat *and* the data source connected (`g\_Datasources.Framo\_nState`), so a dead session is seen before the heartbeat times out. Only the pump variables are mapped, not the server's standard `Server` object. Start, Stop and Reset are events acted on at the rising edge[OI4]. The PLC holds Reset TRUE from the operator's command until it reads the pump out of LOCKDOWN, at most 2×`ResetExpectedTime`; then it withdraws Reset and warns *Failed to reset*.
- **PLC ↔ InletRIO n:** Modbus TCP, function code 23, every 100 ms.
- **HMI ↔ PLC:** OPC UA (PLC is the server).

5 Non-functional requirements

5.1 Control law and tuning

LIC01 is a PI controller (CODESYS `Util.PID` wrapped in `PIDController`):

$$T(t) = K_p\Big(e(t) + \frac{1}{T_i}\int e\,dt\Big) + Y_{\text{offset}}, \qquad e = \text{SP} - \Delta H_{\text{avg}},$$

the total speed of the counted pumps, limited to the others' speed $+|R|\times$(40–130) rpm and shared as in Section 3.2 (`FB.Inlet.PenLevelControl`); while the output is at a limit Util.PID adapts the integral to the limited output (anti-windup). Manual→auto gives `RESET` for one cycle (clears the integral and applies K_p, T_i) with $Y_{\text{offset}} = Y_{\text{prev}} - K_p e$, so the first automatic output is Y_{prev} (bumpless); on the operator's manual→auto the active setpoint starts at the PV, so $e = 0$ and $Y_{\text{offset}} = Y_{\text{prev}}$, and then ramps to the operator's setpoint. Auto→manual (`Pen.LevelController.Status.Manual` rising) sets `Pen.LevelController.Command.ManualOutput` $= Y_{\text{prev}}/N$. In manual Util.PID sets $Y = Y_{\text{manual}}$ without its limits; `PIDController` therefore limits the manual output to `OutputMin`..`OutputMax` (all four controllers), so manual never drives LIC01 below $N \times 40$ rpm. Util.PID applies new K_p, T_i only at a reset: after changing the tuning in auto, give *Auto* once (restart from the current output).

Table 9: LIC01 start values.

Parameter	[PLC]	[FD] Tabell 8	Comment
K_p	1350 total rpm/m	P = 0.5 / 1 / 2 for 1 / 2 / 3 pumps	= the former 450 rpm/m per pump with 3 pumps (≈ [FD] P = 1: 90 rpm over 0.20 m). With the total output one set holds for every pump count (DEC5)
T_i	60 s	600 s	600 s gives a closed loop of about 16 min [DES]; 60 s tested[DEC4]
T_d	0	–	PI
Output	$N\times$40–130 rpm (total)	40–130 rpm	per pump 40–130 rpm
PV scaling	m	0–20 cm	
Margins (worst case O-35, 3 pumps)	PM 76°, GM 13.2 dB, $1/\omega_c$ = 100 s	–	[DES] §7, dead time 2.5θ

Simulator tests (PFC200 against CasRioSim/CasFramoSim, O-35, 3 pumps, K_p 450 per pump, T_i 60 s, before the sharing):

- start-up: pumps reach REMOTE at about 87 rpm, take-over bumpless;
- setpoint step +2 cm: settles in about 100 s, no overshoot;
- one of three pumps lost: the level dips 3.4 cm; the pump rejoining gives 2.7 cm overshoot. There is no feed-forward on the number of pumps, and the integral winds up against 130 rpm while two pumps carry the load.

Baseline (2026-10-22, same rig): pump 3 out ⇒ level 6.0 → 3.0 cm, about 4 min back; pump 3 restart ⇒ 9.1 cm, mostly during its START (its own preset speed while the others still ran fast). **Change** (DEC5): the output is the total speed, shared by the pumps in REMOTE, so a trip or a start is compensated in the same cycle. An I-PD form would avoid the setpoint kick on mode changes.

Live test (2026-10-22, PFC200 10.1.2.41 on `main` after Reset Origin, CAS plant simulator 10.1.2.150, O-35):

- Pen Start from a drained pen (−3.4 cm): Starting → Running when the pumps reached REMOTE; LIC01 tracked the summed speed (260 rpm) and handed over at 85.5 rpm per pump (bumpless); level peak 6.6 cm (0.6 cm overshoot), settled at 6.1 cm in about 2.5 min with three pumps at 74 rpm.
- Pump stopped by the pump count (test values `PumpTargetSpeed` 115 rpm, delay 60 s): the pump with the most running days stopped, the other two took its share in the same cycle; level dip 6.0 → 5.2 cm, back to 6.3 cm; two pumps at about 112 rpm, no flapping.
- Rotation (test value `PumpCycleTime` 2 min): the standby started first and the old pump stopped once the new one was in REMOTE; level 5.1–7.1 cm during each handover (register IN-31). With a short cycle the rotations follow each other, as every running pump has passed it.
- Commands: O_2 Start with the pen Stopped was refused (`CommandRejected`); all pulses were cleared by the PLC in the same scan.
- Not yet tested: pump trip, level and speed deviation alarms, RIO link loss with the new code.

Small setpoints. O-60 and S-75 use a 0.5 cm setpoint, against about 1 mm resolution of the over-height, ±6 mm combined transmitter uncertainty and waves up to $H_s = 2$ m. Whether this is controllable, and with what filter, is open[OI7].

5.2 Cycle times and tasks

Table 10: Tasks and timing ([PLC] Task Configuration of PFC200_Sim).

Task	Cycle	Inlet content
`Task`	20 ms	`Main`: InletSupervisor (LIC01, sequence), InletControl[0..2] (pump control, heartbeat, InletSensors), PenControl (PV); `IO_Processing` (InletRIO masters)
`OPC_Processing`	50 ms	`FramoDataProcessing`: maps Framo_n to/from GVLMain
`DatasourcesTask`	50 ms	OPC UA client (Data Sources) to the PCCs
Modbus TCP poll	100 ms	InletRIO n, FC23

A level reading reaches LIC01 two `Task` cycles after it is read; the speed setpoint reaches the PCC within some tenths of a second plus the OPC UA publishing time. This is small against the level loop's dead time of about 2 s and time constant of 28–40 s [DES]. Task watchdogs are disabled in the test application[OI23].

5.3 Signal resolution and filtering

- **750-455:** value in bits 14..4 (0..2047), bits 3..2 reserved, bits 1..0 diagnostics; 20 mA = `0x7FF0`[DEC2]. On the 0–2 m span one step is 0.98 mm; the over-height, the difference of two readings, moves in steps of about 1 mm. The PLC scales Raw · (Max − Min)/32760 + Min (20 mA reads 99.98 % of span).
- **Transmitter accuracy:** inside ±2 mm (radar), outside ±4 mm (pressure, 0.2 % of an assumed 2 m full scale)[OI18].
- **Filter:** 2nd-order Butterworth per pair, $f_c = 0.05$ Hz, bilinear with pre-warping, sample time = task cycle read from the runtime, LREAL coefficients. A 1 cm step reaches 63 % in 5.6 s with 4 % overshoot; 4–10 s waves are attenuated by about 19 dB [DES]. The filter restarts at the current value when a pair becomes valid.
- The per-transmitter filters (`FilterFc` 0.05 Hz) are for display only.

- **Signal direction:** raw 0 (4 mA) is assumed to be the lowest level; to be read in the transmitters. Inside and outside must agree, or the sign of LIC01 is wrong[OI18].

5.4 Alarm functions

Alarm states are handled by `AlarmManagement` (raised on the condition, acknowledged on the rising edge of the acknowledge bit). [AL] defines no priorities or acknowledgement classes; the column Class is a proposal (A = alarm, operator action; W = warning; N = notification only)[OI21].

A pump out of service raises no alarms: a pump is in service when it takes part in the pump count (`Inlet[i].InletPump.Config.InletPumpEnable`) or is in manual. The communication loss alarm needs the pump in service; the alarms from the PCC's flags (drive, power, speed deviation and the common alarm) also need a live link (`InletPump.Status.CommunicationOK`), so no alarm is raised from values that are no longer read. LIC01's and OIC01's limit and overflow alarms are raised only while the controller is in automatic control (LIC01 mode Auto; OIC01 running, not in manual, with a good O_2 measurement).

Table 11: Alarms of the inlet system ([FD] Tabell 7 and 9, [AL], [PLC]). Per pump n = 1..3 unless noted.

Alarm	Condition	Class	Action	PLC
Communication loss pump n	PCC has not cleared HeartbeatToggle for 10 s	A	UNAVAILABLE; last speed kept in LIC01's total; O_2 line n stops	yes
Start of pump n failed	Start sent and still AVAILABLE after StartupTime (30 s)	A	none	yes
Stop of pump n failed	Stop sent and still in START, REMOTE or LINK LOST after ShutdownTime (30 s)	A	none	yes
Failed to reset pump n	Reset given in LOCKDOWN, no state change within 2×ResetExpectedTime (10 s); clears when the pump leaves LOCKDOWN or on a new Reset	W (auto-ack)	Reset withdrawn	yes
Pump n failed (common)	any of DriveError, DriveCommunication, DriveNotInRemote, MotorTemperature, EmergencyStop, DCUndervoltageError, PumpStopped, UPSSupplyError, EmergencyRunMode	A	shown on info panel	yes[OI10]
Individual Framo alarms	each flag above, own alarm	A	logged	yes
Pump n low / high power	LowPower / HighPower from PCC; not in the common alarm	N (auto-ack)	none, notification only [FD]	yes

Alarm	Condition	Class	Action	PLC
Speed deviation pump n	$\lvert n_{SP} - n \rvert >$ `SpeedDeviationLimit` (10 rpm) for `SpeedDeviationDelay` (30 s), in REMOTE (`Inlet[i].InletPump.Alarm.SpeedDeviation`)	W	none	yes
Vacuum overfill line n	LSH3n for 5 s (configurable)	A	none	no[OI11]
Emergency stop activated	EmergencyStop from PCC (hard-wired TBD)	A	pump stops	via Framo[OI13]
Level deviation high / low	ΔH_{avg} more than `LevelDeviationLimit` (0.02 m) above / below SP for `LevelDeviationDelay` (5 min), LIC01 in auto (`Pen.Alarm.LevelDeviationHigh`, `.LevelDeviationLow`)	A	none	yes
Low level in pen, warning / alarm	$\Delta H_{avg} <$ `LevelWL` 0.03 / `LevelAL` 0.02 m for `LevelAlarmDelay` (10 s), valid measurement (`Pen.Alarm.LevelWL`, `.LevelAL`)	W / A	none (LALL 0.01 m, close the outlet: not implemented)	yes
High level in pen, warning / alarm	$\Delta H_{avg} >$ `LevelWH` 0.08 / `LevelAH` 0.09 m for `LevelAlarmDelay` (`Pen.Alarm.LevelWH`, `.LevelAH`)	W / A	none	yes
High-high level in pen	$\Delta H_{avg} >$ `LevelAHH` 0.10 m for `LevelAlarmDelay` (`Pen.Alarm.LevelAHH`)	A	trip: all inlet pumps stop (also in manual), the pen stops; `Pen.Status.LevelHighTrip` until below `LevelWH`, then the operator starts the pen	yes
Level transmitter fault (LT01–03, LT21–23)	750-455 status bits $\neq$ 00 for 1 s	W	pair drops out of PV	generic quality alarm[OI8]
Level measurement lost (LAL01)	no valid pair (pen)	A	LIC01 at SafeOutput	yes
LIC01 output at limit / overflow	`LimitsActive`, `Overflow` (pen)	W	none	yes
InletRIO n communication error	Modbus `CommOK` FALSE	A	pair n drops out of the PV	yes
InletRIO n field power	XF05 tripped	A	LTs of inlet n lost	yes

5.5 Parameters

All parameters are written from the HMI and retained: `Main.UpdateRetain` keeps them in step with `GVLRetain`. Defaults from `ConfigValues` apply on a fresh download (rev B added fields to

`PenConfigDUT`: download with Reset Origin).

Table 12: Configurable parameters of the inlet system.

Parameter	Default	Unit	Meaning
`Pen.LevelController.Config.Kp`	1350	rpm/m	LIC01 gain on the total speed
`Pen.LevelController.Config.Ti`	60	s	integral time (new value takes effect at the next start or manual→auto)
`Pen.LevelController.Config.Td`	0	s	derivative time
`Pen.LevelController.Config.` `MinOutput` / `MaxOutput`	40 / 130	rpm	speed range per pump (LIC01: $\times N$)
`Pen.LevelController.Config.` `SafeOutput`	60	rpm	speed per pump with no valid level pair
`Pen.LevelController.Config.` `SetpointRamp`	0.01	m/min	active setpoint from the PV to the operator's after manual→auto; 0 = at once
`Pen.Config.LevelDeviationLimit` / `LevelDeviationDelay`	0.02 / 5	m / min	level deviation alarms; limit 0 = off
`Pen.Config.LevelWL/AL/WH/AH/AHH`	0.03 / 0.02 / 0.08 / 0.09 / 0.10	m	level limits; WL..WH is the setpoint range; AHH trips the inlet pumps
`Pen.Config.LevelAlarmDelay`	10	s	delay of the level limits
`Pen.Config.LevelAlarmHysteresis`	0.005	m	a level limit clears only this far back inside
`Pen.Config.PumpTargetSpeed`	85	rpm	stop a pump when the others would run at or below this; 0 = never
`Pen.Config.PumpAddSpeed`	115	rpm	start a pump when the share is at or above this; above `PumpTargetSpeed`; 0 = never
`Pen.Config.PumpRemoveDelay` / `PumpAddDelay`	15 / 2	min	how long the stop / start condition must hold
`Pen.Config.PumpCycleTime`	24	h	time in REMOTE before a standby takes over; 0 = no rotation
`Inlet[i].InletPump.Config.` `InletPumpEnable`	TRUE (also the default)		pump i takes part in the pump count and rotation
`Pen.LevelController.Command.` `Setpoint`	0.06	m	over-height setpoint (O-35)
`LevelInsideConfig[i]`: Enable, MinValue, MaxValue	TRUE, 0, 2.0	m	LT0i in use and range
`LevelInsideConfig[i]`: FilterFc, QualityEnabled	0.05, TRUE	Hz	display filter; quality alarm
`LevelOutsideConfig[i]`: as inside	as inside		LT2i
`OverHeightConfig[i]`: Enable, FilterFc	TRUE, 0.05	Hz	pair in use; cut-off of the pair's filter (0 = off)
`Inlet[i].InletPump.Config.` `HeartbeatInterval`	1	s	PLC sets the bit this long after the PCC cleared it
`Inlet[i].InletPump.Config.` `HeartbeatTimeout`	10 (DUT 5)	s	link-loss detection
`Inlet[i].InletPump.Config.` `StartupTime` / `ShutdownTime`	30 / 30	s	start / stop failed alarms

Parameter	Default	Unit	Meaning
`Inlet[i].InletPump.Config.SpeedDeviationLimit` / `SpeedDeviationDelay`	10 / 30	rpm / s	speed deviation alarm in REMOTE; limit 0 = off
`Inlet[i].InletPump.Config.ResetExpectedTime`	5	s	expected PCC response to Reset (OPC UA round trip ≈4.6 s); warning *Failed to reset* after 2×
`Inlet[i].RIO.Config`: IP, Port	sim: 10.1.2.150, 5121–5123		coupler address (site plan TBD)
`Inlet[i].InletPump.Config.FramoIP`, `.FramoPort`	sim: 10.1.2.150, 4841–4843		OPC UA endpoint of PCC i (data source `Framo_i`, dynamic configuration); 0.0.0.0 = not configured
`Inlet[i].InletPump.Config.FramoUpdateRate`	500	ms	sampling and publishing interval of the data source's subscription OI8 OI9 OI11 OI14
Not yet defined: L/H level limits and delays, speed deviation limit, vacuum overfill delay (5 s), mode table SP	TBD		

5.6 Interfaces: instruments and actuators

Table 13: Instruments, actuators and PLC functions of the inlet system ([PID], [EL], [DES]). Use: C = control, M = monitor, A = alarm.

Tag	Function	Location	Signal / interface	Range	Resolution	Use
LT01–03	Level inside the pen, at inlet n; VEGAPULS radar OI18	Gooseneck, inlet n	4–20 mA 2-wire, InletRIO n -R1-106 ch1, word 0	0–2 m	0.98 mm; ±2 mm	C, M, A
LT21–23	Sea level outside the pen, at inlet n; submersible pressure probe	Outside the collar, inlet n	4–20 mA 2-wire, InletRIO n ch2, word 1	0–2 m (assumed)	0.98 mm; ±4 mm	C, M, A
LDY01–03	Over-height of pair n, filtered	PLC	computed	0–0.20 m	≈1 mm	C
LY01	Mean over-height of valid pairs (PV)	PLC	computed	0–0.20 m	–	C, A
LIC01	Level controller, PI	PLC	output to SY31–33	40–130 rpm	–	C
LAL01	Level measurement lost	PLC	alarm	–	–	A
PM-31–33	Framo SR2000 inlet pump, rim motor 60 kW	Intake n	via PCCn (ACS880)	40–130 rpm, 20 000 m^3/h	–	actuator
SC31–33	Framo pump controller CX8200 + ACS880-31-105A-3	PCCn	OPC UA server	–	–	–
SY31–33	Speed setpoint to pump n	PLC → PCCn	OPC UA SpeedSetpoint	40–130 rpm	REAL	C

Tag	Function	Location	Signal / interface	Range	Resolution	Use
SI31–33	Actual pump speed	PCC*n*	OPC UA MotorVelocity	0–130 rpm	REAL	C, M
HS31–33	Start / Stop / Reset pump *n*	PLC → PCC*n*	OPC UA events	–	–	C
YI31–33	Pump state	PCC*n*	OPC UA 7 flags	7 states	–	C, M
UA31–33	Pump alarms (11 flags)	PCC*n*	OPC UA alarms	–	–	A
KY31–33	Heartbeat	PLC ↔ PCC*n*	OPC UA BOOL	–	–	C, A
LSL31–33	Vacuum chamber low level (air evacuated)	TV-3*n*	OPC UA (Framo LSL01)	on/off	–	M
LSH31–33	Vacuum chamber high level (overfill)	TV-3*n*	OPC UA (Framo LSH02)	on/off	–	A[OI11]
PT31–33	Compressed air pressure to ejector	Air line, inlet *n*	OPC UA analog	Framo range	REAL	M
XV31A–33A	Air valve to ejector EJ-3*n*	Inlet *n*	status OPC UA (V001?)	open/closed	–	M[OI11]
XV31B–33B	Vacuum break (vent) valve	TV-3*n*	status OPC UA (V002?)	open/closed	–	M[OI11]
E-stop PCC*n*	Local emergency stop, STO	PCC*n* door	OPC UA EmergencyStop	on/off	–	A[OI13]
YA30 (n)	24 V field breaker XF05 tripped (LT supply)	RIO *n* cabinet	InletRIO *n* DI word 4.0	on/off	–	A

5.7 Interfaces: OPC UA signals to and from the Framo controllers

Per pump *n* the PLC uses the Data Source `Framo_n`; the GVL side is `GVLMain.Inlet[n-1].InletPump`. Direction is seen from the PLC (R = read, W = written).

Table 14: OPC UA signal list, PLC ↔ Framo PCC*n* ([FRAMO] rev. B, [FD] Tabell 6, [PLC] FramoDataProcessing).

Signal (Data Source name)	Type	Dir.	Use in PLC	Remark
MotorVelocity (+ ValueRange Low/High)	AnalogValue	R	tracking, bumpless transfer, speed indication	range written by Framo?[OI19]
MotorFrequency, MotorCurrent, MotorPower, MotorVoltage, MotorTorque, PowerConsumptionCounter, DCLinkVoltage, VSDHeatSinkTemperature, VSDOperatingDays	AnalogValue	R	info panel	[FRAMO]: PowerConsumtionCounter
MotorRunningDays	AnalogValue	R	info panel; run-time order of pumps (`SortSequence`) for the pump count and rotation	

Signal (Data Source name)	Type	Dir.	Use in PLC	Remark
DriveError, DriveCommunication, DriveNotInRemote, MotorTemperature, EmergencyStop, DCUnderVoltageError, PumpStopped, UPSSupplyError, EmergencyRunMode	Alarm	R	own alarm each; common alarm	
LowPower, HighPower	Alarm	R	notification (auto-ack), not in the common alarm	
PumpStateUnavailable, ... Available, Start, Stop, Remote, Hold, Lockdown	Event	R	`InletPump.State` (first TRUE flag)	
Start, Stop, Reset	Event	W	commands, rising edge	OI3 OI4
SpeedSetpoint	AnalogValue	W	LIC01 output in REMOTE, else actual speed or 0	[FRAMO], [FD]: Velocity-Setpoint OI19
HeartbeatToggle	BOOL	R/W	PLC sets, PCC clears	OI1
HeartbeatInterval, HeartbeatElapsed	TIME	–	not mapped	OI1
PressureCompressedAir, LevelSwitchLSL01, LevelSwitchLSH02, StatusV001, StatusV002	Analog / BOOL	R	vacuum system status	not in [FRAMO] / [FIL] OI11

5.8 HMI operator functions

6 Open issues and decisions

6.1 Open issues

Each item names the sources that disagree and, where there is one, a proposal. “Framo” items need agreement with the supplier.

OI1 Heartbeat baseline. PLC: one shared BOOL, PLC sets / PCC clears, neither side waits after connect, interval 1 s, timeout 10 s (DUT default 5 s). [FRAMO] rev. B: HeartbeatToggle with HeartbeatInterval and HeartbeatElapsed. [FD]: INT counter 0–32767. Agree one with Framo, incl. the PCC’s own timeout and whether the PLC writes Interval.

OI2 HOLD → REMOTE. Closed 2026-11-05 by [FRAMO-FD] rev. 04: no HOLD; on link loss the pump runs at last setpoint × 0.8 and follows the setpoint again when the link is back, without Reset. Still open with Framo (questions of 2026-11-05): the interface list for rev. 04 (state flags, stand-alone, Remote and Inhibit commands), what the PCC reports while on the reduced setpoint, and what happens on a link loss during start or stop.

OI3 LOCKDOWN exit and Reset. [FD]: operator Reset after the fault is fixed. [FRAMO]: LOCKDOWN ends with “sequence end” to UNAVAILABLE, no Reset. What Reset does (all alarms? edge?) is not described (Framo review comment).

OI4 Stop and command semantics. [FRAMO] rev. B accepts Stop only in REMOTE (not in START). The PLC sends Stop in any state and raises Stop failed for START, REMOTE and LINK LOST. Are Start/Stop/Reset edge- or level-triggered?

Table 15: Operator functions for the inlet system (PLC variables in [PDS] Table “Commands”).

Function	PLC variable	Note
Pen Start / Pen Stop	`Pen.Command.Start`, `.Stop`; shows `Pen.Status.State`, `.Ready`, `.CommandRejected`	pulses; Start only when Ready; starts the enabled pumps in auto
Oxygen Start / Stop	`Pen.Command.OxygenStart`, `.OxygenStop`; shows `Pen.Status.OxygenOn`	pulses; enabled lines follow their inlet pumps
Pump count and rotation	`Inlet[i].InletPump.Config.InletPumpEnable`; shows `Pen.InletSupervisor.PumpsWanted`, `.PumpRotationActive`	take a pump out of the count without manual
Level setpoint	`Pen.LevelController.Command.Setpoint`	m; from the mode in future[OI14]
LIC01 auto / manual, manual output	`Pen.LevelController.Command.Manual`, `.Auto` (pulses; mode in `.Status.Manual`), `.ManualOutput`	rpm per pump
Pump auto / manual	`Inlet[i].InletPump.Command.Auto`, `.Manual` (pulses; mode in `.Status.Manual`)	default auto
Pump start / stop / speed in manual	`Inlet[i].InletPump.Command.Start`, `.Stop`, `.Setpoint`	speed used in REMOTE; Start / Stop written in auto are cleared, they do not act at the switch to manual
Reset pump (LOCKDOWN)	`Inlet[i].InletPump.Command.Reset`	pulse, accepted only in LOCKDOWN; the PLC clears it and holds the PCC’s Reset until it sees the state change. HMI button per pump[OI24]
Enable/disable transmitter or pair	`...LevelInsidePen.Config.Enable`, `...PenOverHeight.Config.Enable`	takes a pair out of the PV
Info panel per pump	`Inlet[i].InletPump.Input.*`, `.State`	[FD] Figur 3
Alarm list and acknowledgement	`*.Alarm.*`, `*.AlarmAck.*`	classes TBD[OI21]
Parameters (Table 12)	`GVLMain.Pen.Config`, `InletConfig`, ...	access level TBD

OI5 LIC01 parameter sets. *Closed in rev B* by DEC5: with the total output the loop gain does not depend on the pump count; one set (K_p 1350 total rpm/m, T_i 60 s). [FD] Tabell 8 to be updated.

OI6 Pump-count transients. *Closed in rev B* by DEC5 (sharing of the total); to be confirmed in the simulator. I-PD against setpoint kick still to consider.

OI7 0.5 cm setpoint (O-60, S-75). Controllable with ≈1 mm resolution, ±6 mm uncertainty and waves? Decide the setpoint, filter and acceptance criterion.

OI8 Pen level alarms. Deviation alarms high / low implemented (DEC8; 0.02 m, 5 min, to be confirmed). Low and high level alarms of [FD] Tabell 9 / [AL] are not in the PLC; limits and delays TBD. The per-transmitter fault exists only as the generic quality alarm; map it to the [AL] text per tag.

OI9 Speed deviation alarm (in REMOTE) of [FD] Tabell 7: implemented in rev B (10 rpm, 30 s, to be confirmed against Framo’s speed response).

OI10 LowPower / HighPower. *Closed in rev B*: removed from `PumpCommonError`; their own alarms are notifications without acknowledge, as [FD].

OI11 **Vacuum system.** Alarms not implemented (`VacuumSystemAlarmDUT` is empty; LSH overfill with 5 s delay). The vacuum signals are not in [FRAMO] or [FIL]. Tag mapping to confirm: Framo LSL01/LSH02/V001/V002 vs [PID] LSL3n/LSH3n/XV3nA/B vs [FD] LSL31/LSH32/XV31/XV32.

OI12 **Flow figures.** [FD] Tabell 10 O-35 = 34 000 m^3/h; [DES] maximum 27 000 m^3/h; 3 pumps × 20 000 m^3/h rated. Framo comment: "26 000 ok for emergency mode?". Align.

OI13 **Emergency stop.** Do the Framo E-stops (in series, [EL] 874.001) stop all three lines? How is oxygenation stopped on E-stop through the Heimdall PLC (no E-stop input wired today)? [FD] alarm "Nødstopp aktivert" is HOLD.

OI14 **Setpoint from operating mode.** [FD]: LIC01 setpoint chosen from the mode, with manual override. PLC: operator-entered setpoint only; the outlet mode table has damper positions only.

OI15 **Number of pumps and rotation.** Implemented in rev B (DEC7, Section 3.4): stop and start by the share against `PumpTargetSpeed` / `PumpAddSpeed`, rotation by run time after `PumpCycleTime`, next pump when a start fails. Still open: [FD]'s warning when a pump has been idle too long; thresholds, delays and cycle time to be confirmed with Framo (start and stop wear, minimum run time); the level swing of about ±1 cm during a rotation handover (live test 2026-10-22, register IN-31).

OI16 **Auto-restart and start order.** [FD] Tabell 2–3: restart after power failure in a sequence (2a/2b/2c). PLC: pen start latch not retained; Pen Start sends Start to all pumps at once.

OI17 **InletRIO link loss.** *Closed in rev B* by DEC6: the pair is invalid while its RIO has no communication (tested in the simulator 2026-10-22).

OI18 **Level transmitters.** Signal direction (4 mA = lowest level?) to be read in the sensors; inside model VEGAPULS C21 ([DES]) vs C22 ([PID] loop sheet); outside type and 2 m full scale assumed.

OI19 **Framo interface details.** Names differ (SpeedSetpoint vs VelocitySetpoint, DCUnderVoltageError, PowerConsumptionCounter); is the MotorVelocity range written by Framo? Hardware CX8190 ([FRAMO]) vs CX8200 ([EL]); pump SX1000 ([EL]) vs SR2000 ([FD]).

OI20 **Interlocks from other systems.** 2oo3 LS7nmHH trip of PM-3n (I2) and "group start only with the outlet open" (I3) are not in the PLC.

OI21 **Alarm classes.** [AL] has no priority or acknowledgement class; define them.

OI22 **Framo dynamics and preset speeds.** Speed response T_p (10 s assumed), START preset speed (88 rpm assumed) and ramp rates: to be given by Framo (the link-loss factor 0.8 is in [FRAMO-FD] rev. 04).

OI23 **Platform.** Task watchdogs disabled in the test application; IP plan and network drawing missing ([PID] C3).

OI24 **Operator Reset on the HMI.** A Reset button per inlet pump (enabled in LOCKDOWN, with confirmation) is in review for the test HMI (heimdall-cas-hmi); the production HMI needs the same function.

6.2 Decisions

DEC1 2026-10-15 (PLC, in review): PV = mean of the per-pair filtered ΔH_i over the valid pairs; a lost transmitter drops its pair ([FD] §3.2.2).

DEC2 2026-10-07: 750-455 word layout confirmed by WAGO (value in bits 14..4, 0..2047).

DEC3 2026-10-15 (PLC, in review): no valid pair ⇒ alarm LevelMeasurementLost and LIC01 in manual at SafeOutput (60 rpm), bumpless return.

DEC4 2026-10-13: LIC01 start value T_i = 60 s instead of 600 s, from the design basis and the simulator test; to be confirmed at commissioning.

DEC5 2026-10-22: LIC01's output is the total speed of the pumps in level control; the pumps in REMOTE share it after the actual speed of the others in level control. K_p 450 per pump → 1350 total; per-pump limits, `SafeOutput` and manual output scaled by N (IN-05, IN-07).

DEC6 2026-10-22: InletRIO n link lost ⇒ both transmitters of inlet n bad, its pair drops out of the PV (IN-19).

DEC7 2026-10-22: automatic pump count and rotation (Section 3.4): candidates = Enable, auto, not UNAVAILABLE / LOCKDOWN; start all at Pen Start, then reduce; settings retained in `Pen.Config` (IN-15).

DEC8 2026-10-22: level deviation alarms high / low with LIC01 in auto, limit and delay retained in `Pen.Config` (IN-09).

DEC9 2026-10-23 (Sigurd): when a pump goes to HOLD or its link is lost, LIC01 keeps controlling the level with the pumps it can set, instead of freezing ([FD] Tabell 5, [FRAMO] rev B): the three PCCs do not know each other's state. The HOLD pump counts in the total at its own speed (since 2026-11-05: a LINK LOST pump at last setpoint × 0.8). Implemented as rules R1/R2 (Section 3.2) in the split of `InletSupervisor` (IN-33).